

ALGORITHM:

1. Start the program.
2. Display the menu with four choices.
3. Read the user's choice.
4. Use switch-case:
 - Choice 1: Read a string and count spaces, digits, and special characters.
 - Choice 2: Read a string and a character. Traverse the string and count how many times the character occurs.
 - Choice 3: Read a string. Convert uppercase letters to lowercase and lowercase letters to uppercase.
 - Choice 4: Exit the program.
5. For an invalid choice, display an error message.
6. Repeat the menu until choice 4 is selected.
7. Stop.

C PROGRAM:

```
#include <stdio.h>
```

```
void countCharacters();
```

```
void frequency();
```

```
void toggleCase();
```

```
int main()
```

```
{
```

```
    int choice;
```

```
    do
```

```
    {
```

```
        printf("\n----- MENU ----- \n");
```

```
printf("1. Count spaces, digits and special characters\n");  
printf("2. Count frequency of a character\n");  
printf("3. Toggle case of each character\n");  
printf("4. Exit\n");
```

```
printf("Enter your choice: ");  
scanf("%d", &choice);  
getchar(); // Removes newline from input buffer
```

```
switch(choice)  
{  
    case 1:  
        countCharacters();  
        break;  
  
    case 2:  
        frequency();  
        break;  
  
    case 3:  
        toggleCase();  
        break;  
  
    case 4:  
        printf("Exiting program...\n");  
        break;  
  
    default:
```

```

        printf("Invalid choice! Please enter 1 to 4.\n");
    }

} while(choice != 4);

return 0;
}

/* Function to count spaces, digits and special characters */
void countCharacters()
{
    char str[100];
    int i, spaces = 0, digits = 0, special = 0;

    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    for(i = 0; str[i] != '\0'; i++)
    {
        if(str[i] == ' ')
        {
            spaces++;
        }
        else if(str[i] >= '0' && str[i] <= '9')
        {
            digits++;
        }
    }
}

```

```

        else if(!((str[i] >= 'A' && str[i] <= 'Z') ||
            (str[i] >= 'a' && str[i] <= 'z')) &&
            str[i] != '\n')
        {
            special++;
        }
    }

    printf("Number of spaces = %d\n", spaces);
    printf("Number of digits = %d\n", digits);
    printf("Number of special characters = %d\n", special);
}

```

/* Function to count frequency of a given character */

```

void frequency()
{
    char str[100], ch;
    int i, count = 0;

    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    printf("Enter the character to search: ");
    scanf("%c", &ch);
    getchar();

    for(i = 0; str[i] != '\0'; i++)

```

```

{
    if(str[i] == ch)
    {
        count++;
    }
}

printf("Frequency of '%c' = %d\n", ch, count);
}

```

/* Function to toggle the case of characters */

void toggleCase()

```

{
    char str[100];
    int i;

    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    for(i = 0; str[i] != '\0'; i++)
    {
        if(str[i] >= 'A' && str[i] <= 'Z')
        {
            str[i] = str[i] + 32;
        }
        else if(str[i] >= 'a' && str[i] <= 'z')
        {

```

```

        str[i] = str[i] - 32;
    }
}

printf("String after toggling case: %s", str);
}

```

```

#include <stdio.h>

void countCharacters();
void frequency();
void toggleCase();

int main()
{
    int choice;

    do
    {
        printf("\n—— MENU ——\n");
        printf("1. Count spaces, digits and special characters\n");
        printf("2. Count frequency of a character\n");
        printf("3. Toggle case of each character\n");
        printf("4. Exit\n");

        printf("Enter your choice: ");
        scanf("%d", &choice);
        getchar(); // Removes newline from input buffer

        switch(choice)
        {
            case 1:
                countCharacters();
                break;

            case 2:
                frequency();
                break;

            case 3:
                toggleCase();
                break;

            case 4:
                printf("Exiting program ... \n");
                break;

            default:
                printf("Invalid choice! Please enter 1 to 4.\n");
        }
    } while(choice != 4);

    return 0;
}

```

```

/* Function to count spaces, digits and special characters */
void countCharacters()
{
    char str[100];
    int i, spaces = 0, digits = 0, special = 0;

    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    for(i = 0; str[i] != '\0'; i++)
    {
        if(str[i] == ' ')
        {
            spaces++;
        }
        else if(str[i] >= '0' && str[i] <= '9')
        {
            digits++;
        }
        else if(!((str[i] >= 'A' && str[i] <= 'Z') ||
            (str[i] >= 'a' && str[i] <= 'z')) &&
            str[i] != '\n')
        {
            special++;
        }
    }

    printf("Number of spaces = %d\n", spaces);
    printf("Number of digits = %d\n", digits);
    printf("Number of special characters = %d\n", special);
}

/* Function to count frequency of a given character */
void frequency()
{
    char str[100], ch;
    int i, count = 0;

    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    printf("Enter the character to search: ");
    scanf("%c", &ch);
    getchar();

    for(i = 0; str[i] != '\0'; i++)
    {

```

```

int i, count = 0;

printf("Enter a string: ");
fgets(str, sizeof(str), stdin);

printf("Enter the character to search: ");
scanf("%c", &ch);
getchar();

for(i = 0; str[i] != '\0'; i++)
{
    if(str[i] == ch)
    {
        count++;
    }
}

printf("Frequency of '%c' = %d\n", ch, count);
}

/* Function to toggle the case of characters */
void toggleCase()
{
    char str[100];
    int i;

    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    for(i = 0; str[i] != '\0'; i++)
    {
        if(str[i] >= 'A' && str[i] <= 'Z')
        {
            str[i] = str[i] + 32;
        }
        else if(str[i] >= 'a' && str[i] <= 'z')
        {
            str[i] = str[i] - 32;
        }
    }

    printf("String after toggling case: %s", str);
}

```

TEST CASE TABLE:

TEST CASE	CHOICE	INPUT	EXPECTED RESULT
1	1	Hello 123!	Spaces = 1, Digits = 3, Special = 1
2	1	C@SE 2026#	Spaces = 1, Digits = 4, Special = 2
3	2	String: banana, Character: a	Frequency of a = 3
4	2	String: Hello, Character: z	Frequency of z = 0
5	3	Hello World	hELLO wORLD
6	3	CSE123@AbC	cse123@aBc
7	4	--	Program exits
8	6	--	Invalid choice message

